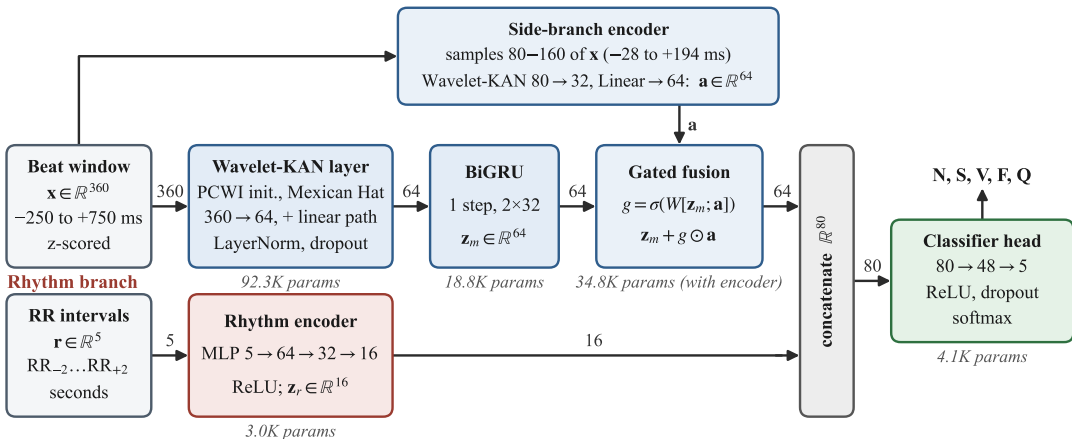


Morphology branch



Training only: class-balanced sampling with unweighted cross-entropy (every retained checkpoint from epochs ≤ 25); augmentation of non-N beats; AdamW; checkpoint selection on DS1-validation Macro-F1.